Big Data Analytics Week 3 Lab: Introduction to Sparklyr ANSWERS

Dom Richards

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Setup and Data Loading

Installing and Loading Required Packages

```
# Install packages if needed
if (!require("sparklyr", quietly = TRUE)) install.packages("sparklyr")
if (!require("dplyr", quietly = TRUE)) install.packages("dplyr")
if (!require("nycflights13", quietly = TRUE)) install.packages("nycflights13")

# Load libraries
library(sparklyr)
library(dplyr)
library(nycflights13)
```

Connecting to Spark

```
# Connect to local Spark instance
sc <- spark_connect(master = "local[*]")
```

Loading the nycflights13 Dataset

The **nycflights13** dataset contains information about all 336,776 flights that departed from New York City airports (JFK, LGA, EWR) in 2013.

```
# Copy data to Spark
flights_tbl <- copy_to(sc, nycflights13::flights, "flights", overwrite = TRUE)
airlines_tbl <- copy_to(sc, nycflights13::airlines, "airlines", overwrite = TRUE)
airports_tbl <- copy_to(sc, nycflights13::airports, "airports", overwrite = TRUE)
planes_tbl <- copy_to(sc, nycflights13::planes, "planes", overwrite = TRUE)
weather_tbl <- copy_to(sc, nycflights13::weather, "weather", overwrite = TRUE)

# Check the data
flights_tbl
```

```
## # Source:   table<`flights`> [?? x 19]
## # Database: spark_connection
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013     1     1      517            515         2      830            819
##  2  2013     1     1      533            529         4      850            830
```

```
##  3  2013     1     1      542            540         2      923            850
##  4  2013     1     1      544            545        -1     1004           1022
##  5  2013     1     1      554            600        -6      812            837
##  6  2013     1     1      554            558        -4      740            728
##  7  2013     1     1      555            600        -5      913            854
##  8  2013     1     1      557            600        -3      709            723
##  9  2013     1     1      557            600        -3      838            846
## 10  2013     1     1      558            600        -2      753            745
## # i more rows
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

Exercise 1: How many rows are in the flights table? Use `count()` and `collect()` to find out.

```
flights_tbl %>%
    count() %>%
    collect()
```

```
## # A tibble: 1 x 1
##        n
##    <dbl>
## 1 336776
```

Selection: filter() {σ}

The `filter()` verb selects rows that satisfy given conditions. It corresponds to the relational algebra selection operator σ.

Exercise 2: Filter the flights to find all United Airlines (carrier code "UA") flights in December that were delayed by more than 2 hours (120 minutes). Print the first 10 rows.

```
flights_tbl %>%
    filter(carrier == "UA", month == 12, dep_delay > 120) %>%
    head(10) %>%
    collect()
```

```
## # A tibble: 10 x 19
##     year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##    <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
##  1  2013    12     2     1154            820       214     1522           1155
##  2  2013    12     2     1644           1310       214     1928           1557
##  3  2013    12     2     1823           1345       278     2123           1640
##  4  2013    12     3     1424           1114       190     1713           1347
##  5  2013    12     4     1944           1630       194     2228           1915
##  6  2013    12     5      954            730       144     1310           1018
##  7  2013    12     5     1355           1145       130     1741           1509
##  8  2013    12     5     1407           1134       153     1643           1428
##  9  2013    12     5     1416           1009       247     1720           1319
## 10  2013    12     5     1544           1259       165     1832           1605
## # i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
## #   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
## #   hour <dbl>, minute <dbl>, time_hour <dttm>
```

Projection: select() {π}

The `select()` verb chooses specific columns from a table. It corresponds to the relational algebra projection operator π.

Exercise 4: Select only the time-related columns (year, month, day, hour, minute) and the distance column. Show the first 5 rows.

```
flights_tbl %>%
    select(year, month, day, hour, minute, distance) %>%
    head(5) %>%
    collect()
```

```
## # A tibble: 5 x 6
##    year month   day  hour minute distance
##   <int> <int> <int> <dbl>  <dbl>    <dbl>
## 1  2013     1     1     5     15     1400
## 2  2013     1     1     5     29     1416
## 3  2013     1     1     5     40     1089
## 4  2013     1     1     5     45     1576
## 5  2013     1     1     6      0      762
```

Combining filter() and select()

You can chain multiple dplyr operations using the pipe operator %>%:

```
# Find delayed flights from Newark, showing only key columns
delayed_ewr <- flights_tbl %>%
    filter(origin == "EWR", dep_delay > 60) %>%
    select(month, day, carrier, flight, dep_delay, arr_delay) %>%
    arrange(desc(dep_delay))

delayed_ewr %>%
    head(10) %>%
    collect()
```

Exercise 5: Find all flights to Los Angeles (LAX) in summer (June, July, August) and show only the carrier, flight number, departure delay, and arrival delay. Order by departure delay (descending).

```
flights_tbl %>%
    filter(dest == "LAX", month %in% c(6, 7, 8)) %>%
    select(carrier, flight, dep_delay, arr_delay) %>%
    arrange(desc(dep_delay)) %>%
    head(10) %>%
    collect()
```

```
## # A tibble: 10 x 4
##    carrier flight dep_delay arr_delay
##    <chr>    <int>     <dbl>     <dbl>
##  1 VX         411       634        NA
##  2 VX         413       413       420
##  3 VX         411       392       404
##  4 AA         119       362       316
##  5 UA        1266       343       313
##  6 UA        1266       325       269
##  7 DL        2363       319       284
##  8 UA        1665       312       277
```

```
##  9 UA        1175       306       319
## 10 UA         841       306       322
```

Aggregation: group_by() and summarise() {γ}

The `group_by()` and `summarise()` verbs compute aggregate statistics for groups of rows.

Basic Aggregation

```
# Average delays by carrier
carrier_delays <- flights_tbl %>%
    group_by(carrier) %>%
    summarise(
        count = count(),
        avg_dep_delay = mean(dep_delay, na.rm = TRUE),
        avg_arr_delay = mean(arr_delay, na.rm = TRUE),
        max_dep_delay = max(dep_delay, na.rm = TRUE)
    )

carrier_delays %>%
    arrange(desc(avg_dep_delay)) %>%
    collect()
```

Exercise 6: Calculate the total number of flights, average departure delay, and average air time for each destination airport. Sort by total number of flights (descending) and show the top 10 destinations.

```
flights_tbl %>%
    group_by(dest) %>%
    summarise(
        total_flights = count(),
        avg_dep_delay = mean(dep_delay, na.rm = TRUE),
        avg_air_time = mean(air_time, na.rm = TRUE)
    ) %>%
    arrange(desc(total_flights)) %>%
    head(10) %>%
    collect()
```

```
## # A tibble: 10 x 4
##    dest  total_flights avg_dep_delay avg_air_time
##    <chr>         <dbl>         <dbl>        <dbl>
##  1 ORD           17283         13.6         116.
##  2 ATL           17215         12.5         113.
##  3 LAX           16174          9.40        328.
##  4 BOS           15508          8.73         39.0
##  5 MCO           14082         11.3         135.
##  6 CLT           14064          9.22         86.2
##  7 SFO           13331         12.9         346.
##  8 FLL           12055         12.7         152.
##  9 MIA           11728          8.88        153.
## 10 DCA            9705         10.3          45.9
```

Exercise 7: Which month has the worst average arrival delays? Calculate average arrival delay by month.

```
flights_tbl %>%
    group_by(month) %>%
```

```
    summarise(avg_arr_delay = mean(arr_delay, na.rm = TRUE)) %>%
    arrange(desc(avg_arr_delay)) %>%
    collect()
```

```
## # A tibble: 12 x 2
##    month avg_arr_delay
##    <int>         <dbl>
##  1     7        16.7
##  2     6        16.5
##  3    12        14.9
##  4     4        11.2
##  5     1         6.13
##  6     8         6.04
##  7     3         5.81
##  8     2         5.61
##  9     5         3.52
## 10    11         0.461
## 11    10        -0.167
## 12     9        -4.02
```

Joins: Combining Multiple Tables {⋈}

Joins combine data from multiple tables based on common columns.

Inner Join

```
# Add airline names to flight data
flights_with_airlines <- flights_tbl %>%
    inner_join(airlines_tbl, by = "carrier") %>%
    select(year, month, day, carrier, name, flight, origin, dest)

flights_with_airlines %>%
    head() %>%
    collect()
```

Exercise 8: Create a query that joins flights with airlines and shows:

- The airline name
- The number of flights for that airline
- The average departure delay

Order by number of flights (descending)

```
flights_tbl %>%
    inner_join(airlines_tbl, by = "carrier") %>%
    group_by(name) %>%
    summarise(
        num_flights = count(),
        avg_dep_delay = mean(dep_delay, na.rm = TRUE)
    ) %>%
    arrange(desc(num_flights)) %>%
    collect()
```

```
## # A tibble: 16 x 3
##    name                        num_flights avg_dep_delay
```

```
##    <chr>                          <dbl>         <dbl>
##  1 United Air Lines Inc.          58665         12.1
##  2 JetBlue Airways                54635         13.0
##  3 ExpressJet Airlines Inc.       54173         20.0
##  4 Delta Air Lines Inc.           48110          9.26
##  5 American Airlines Inc.         32729          8.59
##  6 Envoy Air                      26397         10.6
##  7 US Airways Inc.                20536          3.78
##  8 Endeavor Air Inc.              18460         16.7
##  9 Southwest Airlines Co.         12275         17.7
## 10 Virgin America                  5162         12.9
## 11 AirTran Airways Corporation     3260         18.7
## 12 Alaska Airlines Inc.             714          5.80
## 13 Frontier Airlines Inc.           685         20.2
## 14 Mesa Airlines Inc.               601         19.0
## 15 Hawaiian Airlines Inc.           342          4.90
## 16 SkyWest Airlines Inc.             32         12.6
```